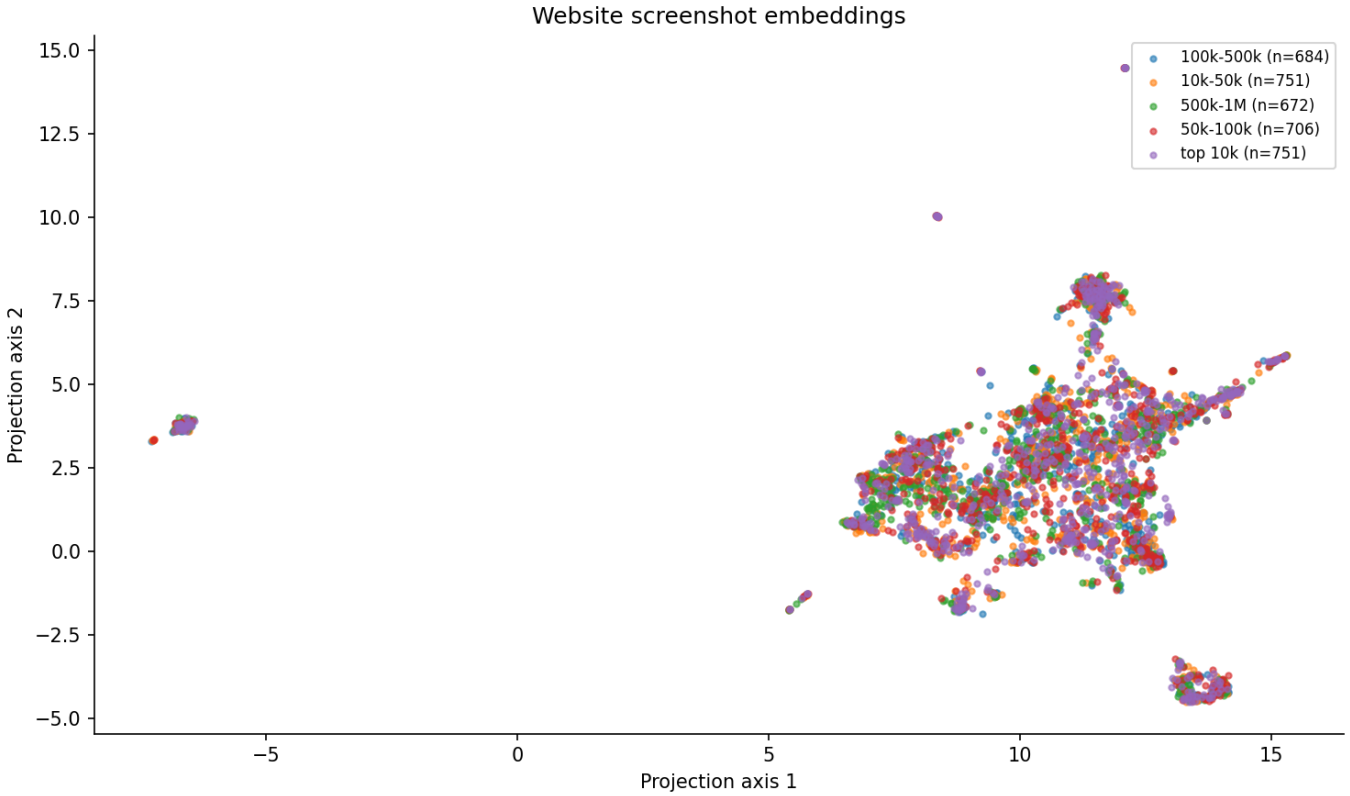
Coverage and the visual map

SUBMISSION READY

5,000/5,000 domains attempted; 3,970 screenshots captured; 3,564 nonblank screenshots embedded. Capture success includes error pages, CAPTCHAs and login screens.

Protocol: Chromium, 1280 x 800 viewport, no scrolling or consent interaction. Pretrained ViT-B-32 / laion2b_s34b_b79k; CPU inference, unit-normalized vectors. OpenCLIP uses a resized center crop: peripheral layout and small text may be lost.



UMAP is exploratory: nearby points suggest relationships but do not establish them. Clusters use KMeans selected from fixed candidates by cosine silhouette (k=12, score=0.100).

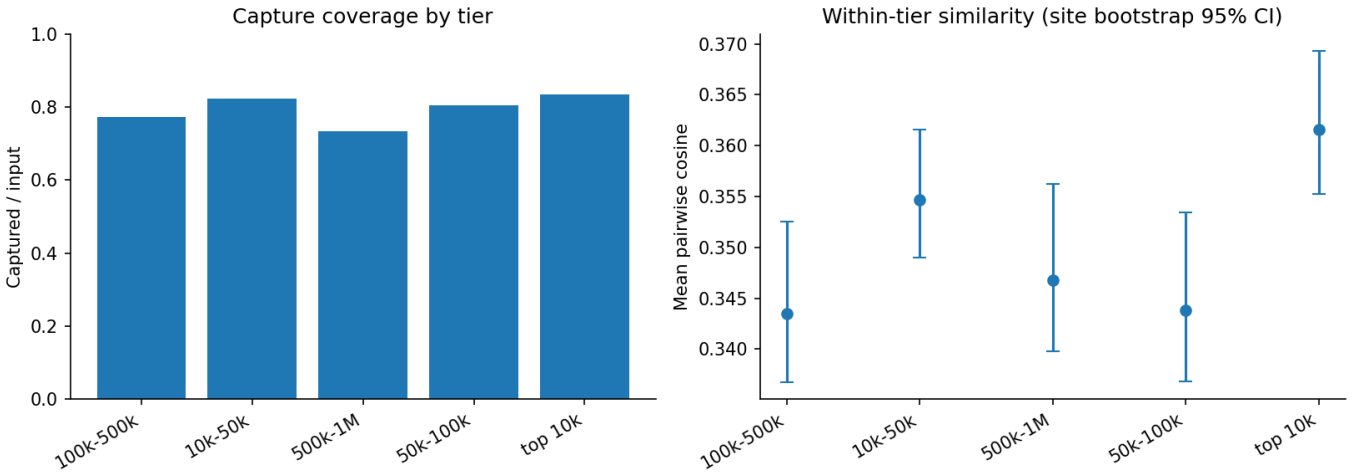
Selection and capture bias: only successfully captured, nonblank images enter the map. Tier denominators include failed visits. These observations describe one browser, network, location and capture time, not a population-wide causal effect of popularity.

Reproduce: python run_pipeline.py. See README for installation, capture settings, manual audit, source references, hashes and submission gates.

1. Do popular websites look different?

SUBMISSION READY

Measured within-tier cosine similarity, with 300 site-level bootstrap resamples. The intervals describe uncertainty within the captured sample; coverage differences can confound tier comparisons.



100k-500k: 774/1000 captured, 684 embedded; mean within-tier cosine 0.343.

10k-50k: 823/1000 captured, 751 embedded; mean within-tier cosine 0.355.

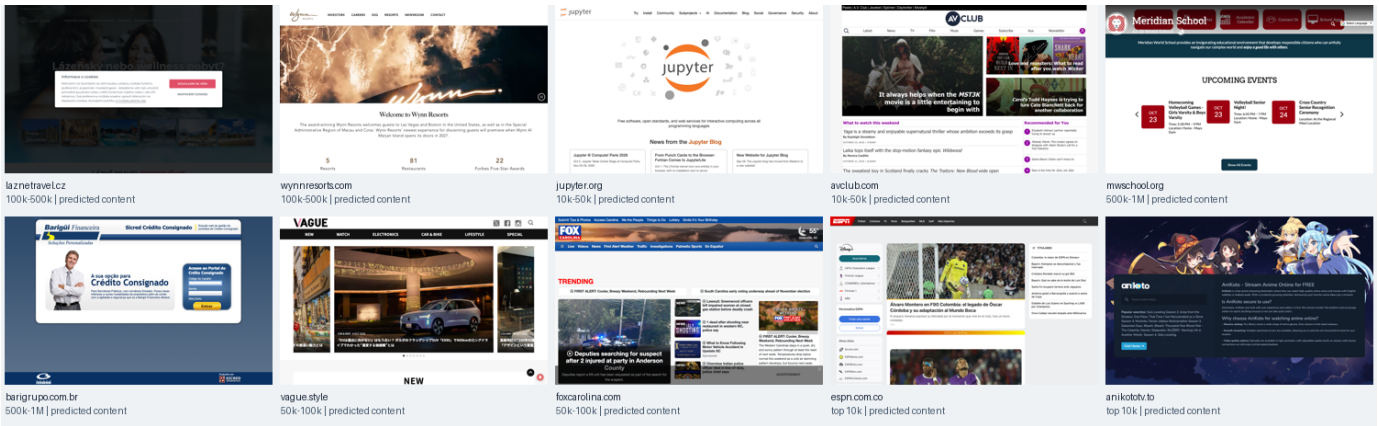
500k-1M: 734/1000 captured, 672 embedded; mean within-tier cosine 0.347.

50k-100k: 804/1000 captured, 706 embedded; mean within-tier cosine 0.344.

top 10k: 835/1000 captured, 751 embedded; mean within-tier cosine 0.362.

Finding 1: top 10k has the highest observed within-tier similarity and 100k-500k the lowest (difference 0.018). This measures visual homogeneity, not aesthetic quality or a monotonic popularity effect.

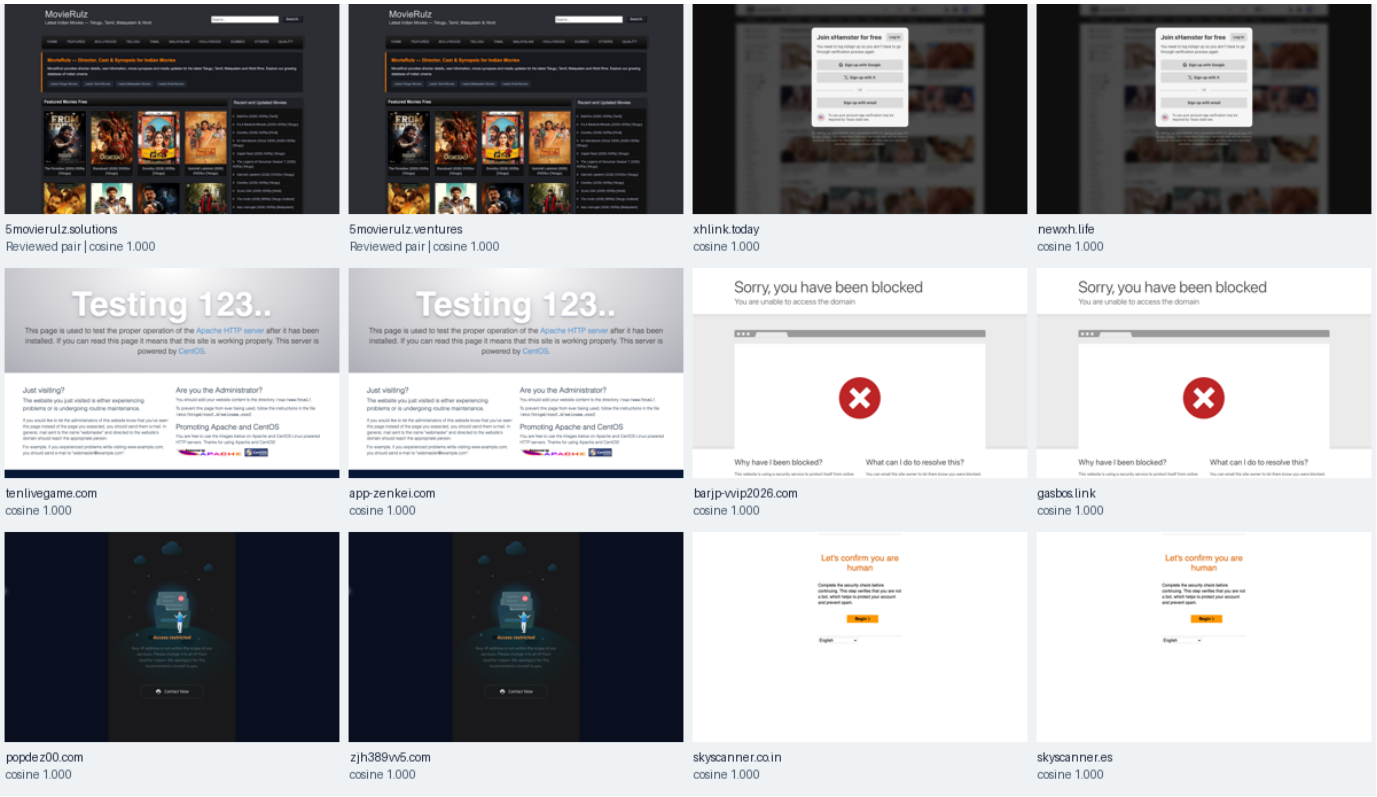
Sensitivity check, predicted content pages only: 100k-500k: n=349, mean=0.336; 10k-50k: n=361, mean=0.356; 500k-1M: n=355, mean=0.335; 50k-100k: n=347, mean=0.336; top 10k: n=355, mean=0.356. These unverified labels do not remove selection bias.



2. Which websites look alike?

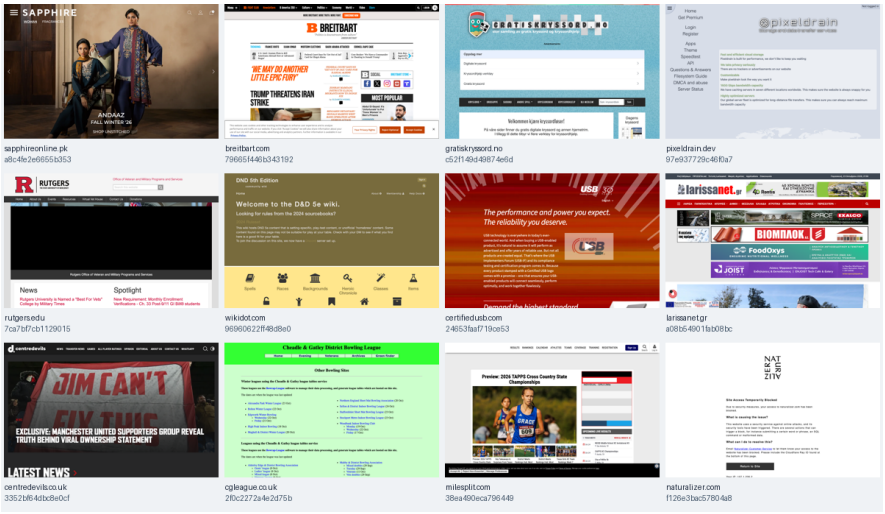
SUBMISSION READY

Finding 2: 409 unique retrieved pairs have cosine at least 0.98. Reviewed examples below include identical captured content and shared access-error templates. The threshold flags candidates; it does not prove common ownership or similar intended website design.



Visual review confirms that 5movierulz.solutions and 5movierulz.ventures display the same dark MovieRulz header, navigation, orange panel and movie-poster grid. Their PNG SHA-256 hashes are identical and cosine is 1.000 (rounded). This establishes an exact captured duplicate, not common ownership.

Outlier evidence: wikidot.com has the lowest nearest-neighbor similarity (0.379). Outliers mean unusual relative to this sample, not unusual on the whole web.

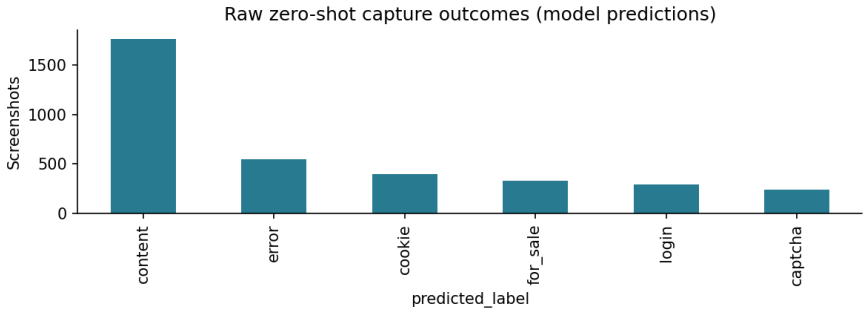


3-4. Capture outcomes and embedding failures

SUBMISSION READY

Finding 3: dominant visible page categories predicted by zero-shot CLIP. Categories compete and may overlap in reality; scores and margins are not calibrated confidence probabilities.

content: 1767 (49.6%); error: 545 (15.3%); cookie: 393 (11.0%); for_sale: 329 (9.2%); login: 292 (8.2%); captcha: 238 (6.7%). Denominator: embedded screenshots, not all domains.



Visual audit by Codex assistant (not independent human): 80/80 reviewed; 78 non-uncertain labels; agreement 62.8%. Precision: content 92% (35/38); login 83% (5/6); captcha 25% (1/4); cookie 0% (0/11); for_sale 0% (0/6); error 62% (8/13). Cookie and for-sale predictions were unreliable in this sample. Raw category counts do not establish true prevalence.

Finding 4: 0912sim.ir and adityanewsnetwork.com have cosine 0.915, but both captures show 403 Forbidden pages: similarity measures an error template, not intended website identity. Separately, CLIP labels the logo-only 000h555.com splash as for_sale although no sale offer is visible. The label set has no splash/loading class. These cases limit semantic conclusions from image vectors and their text-prompt scores.



Limitations: English text prompts, natural-image pretraining, center cropping and small text can obscure website semantics. Access blocks reflect collection outcomes. No CAPTCHA solving or login was attempted. Recaptures can differ because websites and network conditions change.

References: Playwright Python screenshots and Page API (playwright.dev/python/docs); OpenCLIP model and preprocessing (github.com/mlfoundations/open_clip); UMAP documentation (umap-learn.readthedocs.io). Model prompts, package versions, data hashes and settings are archived with the outputs.

Completion gates: see outputs/readiness.json; status shown in the page header.